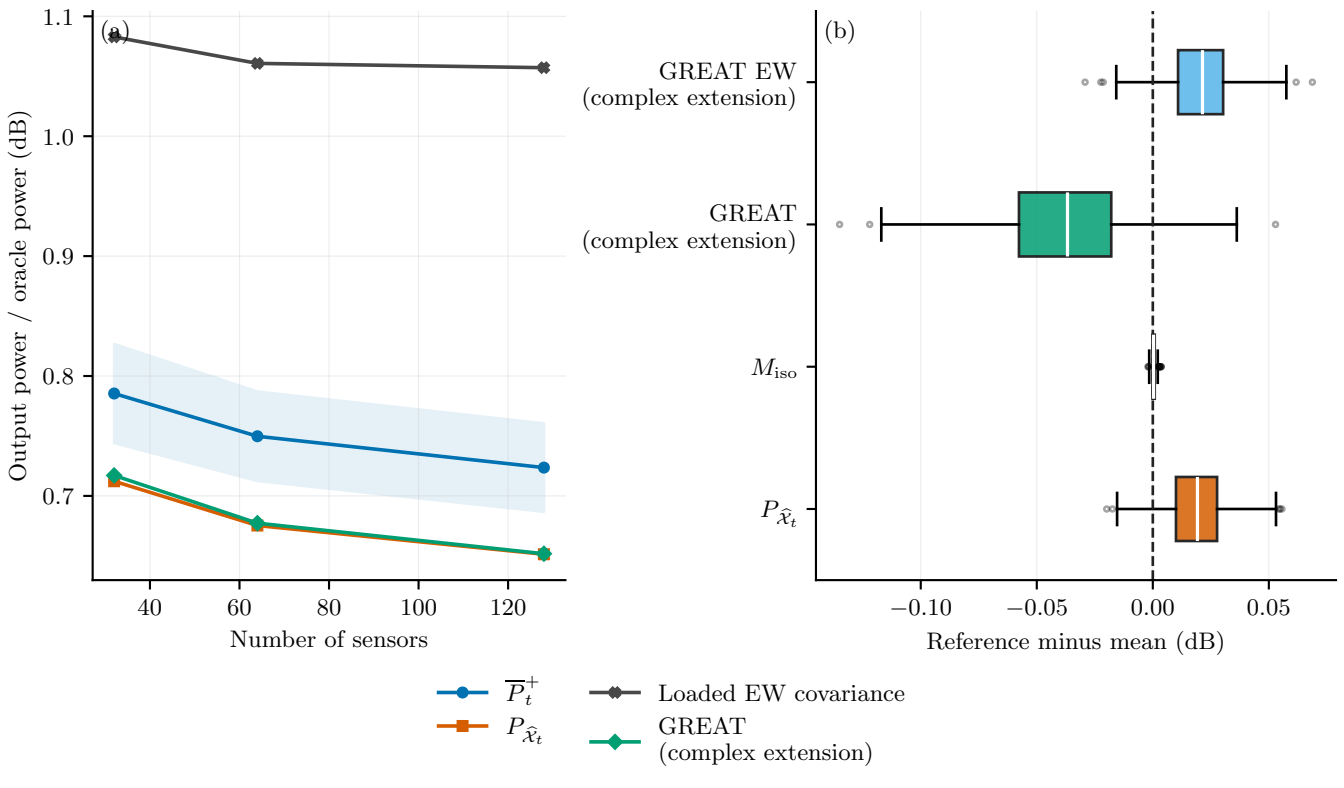
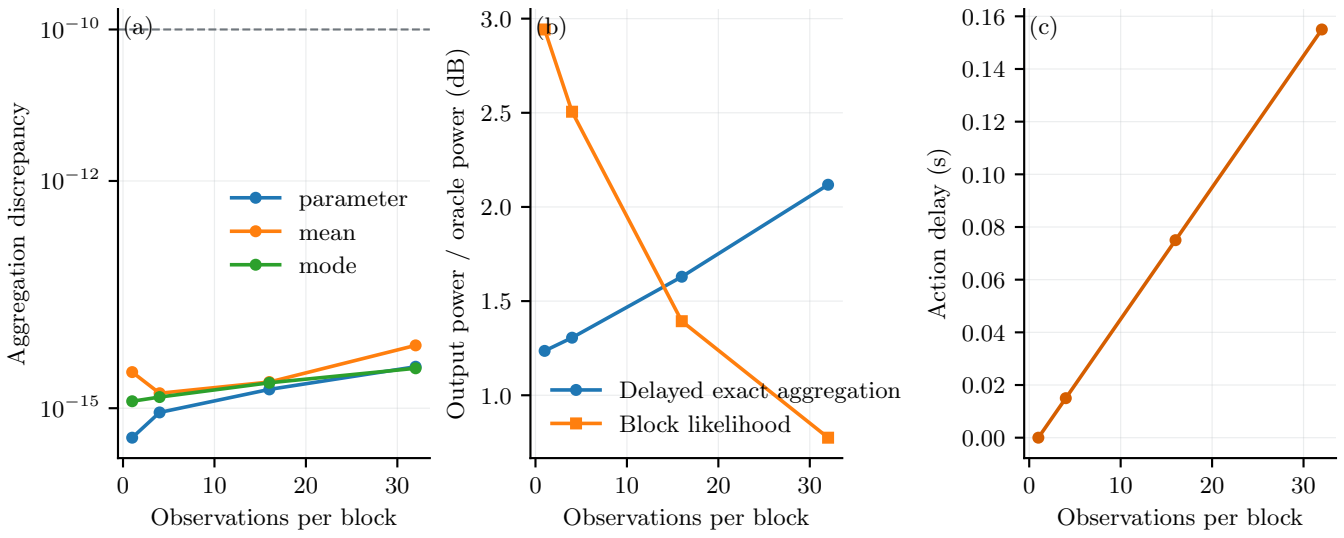
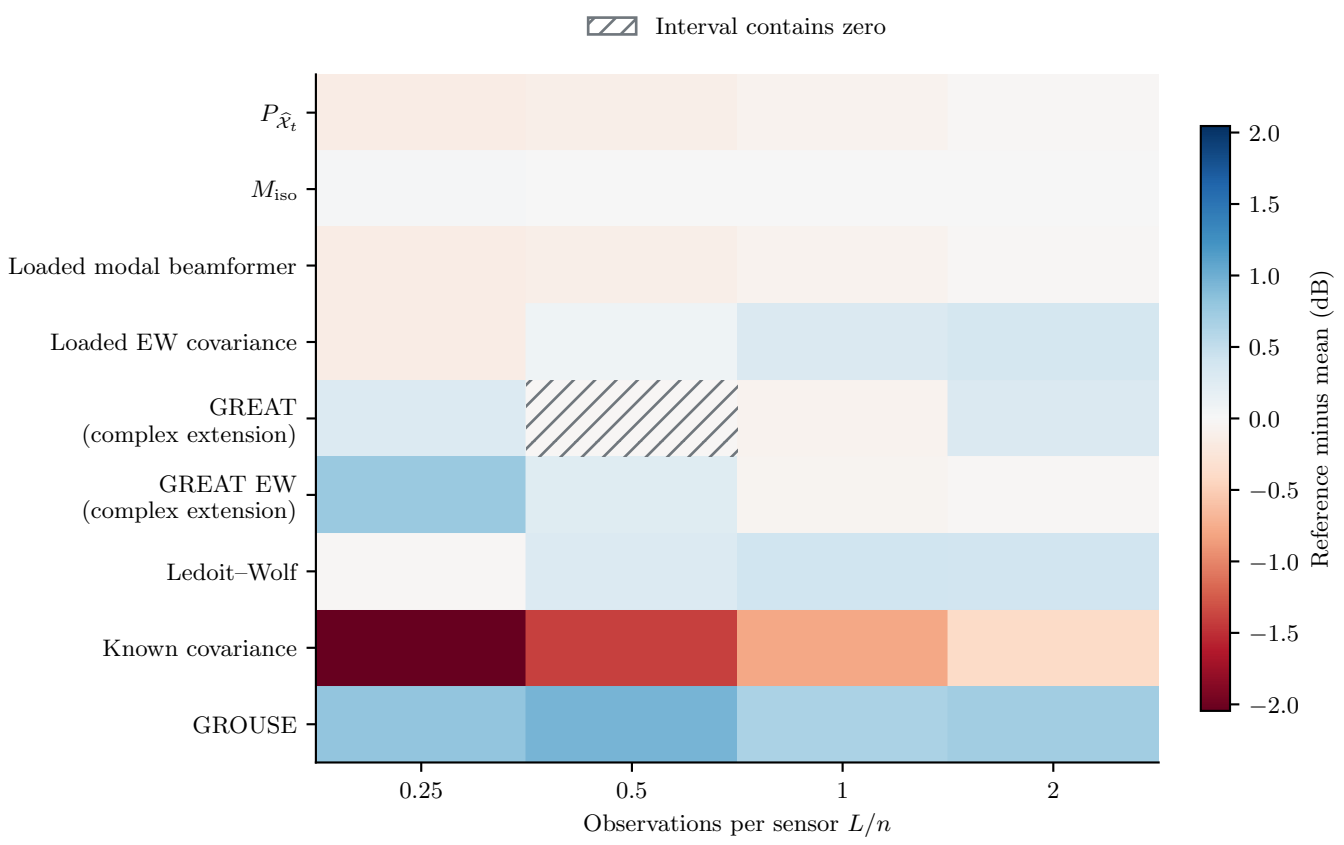
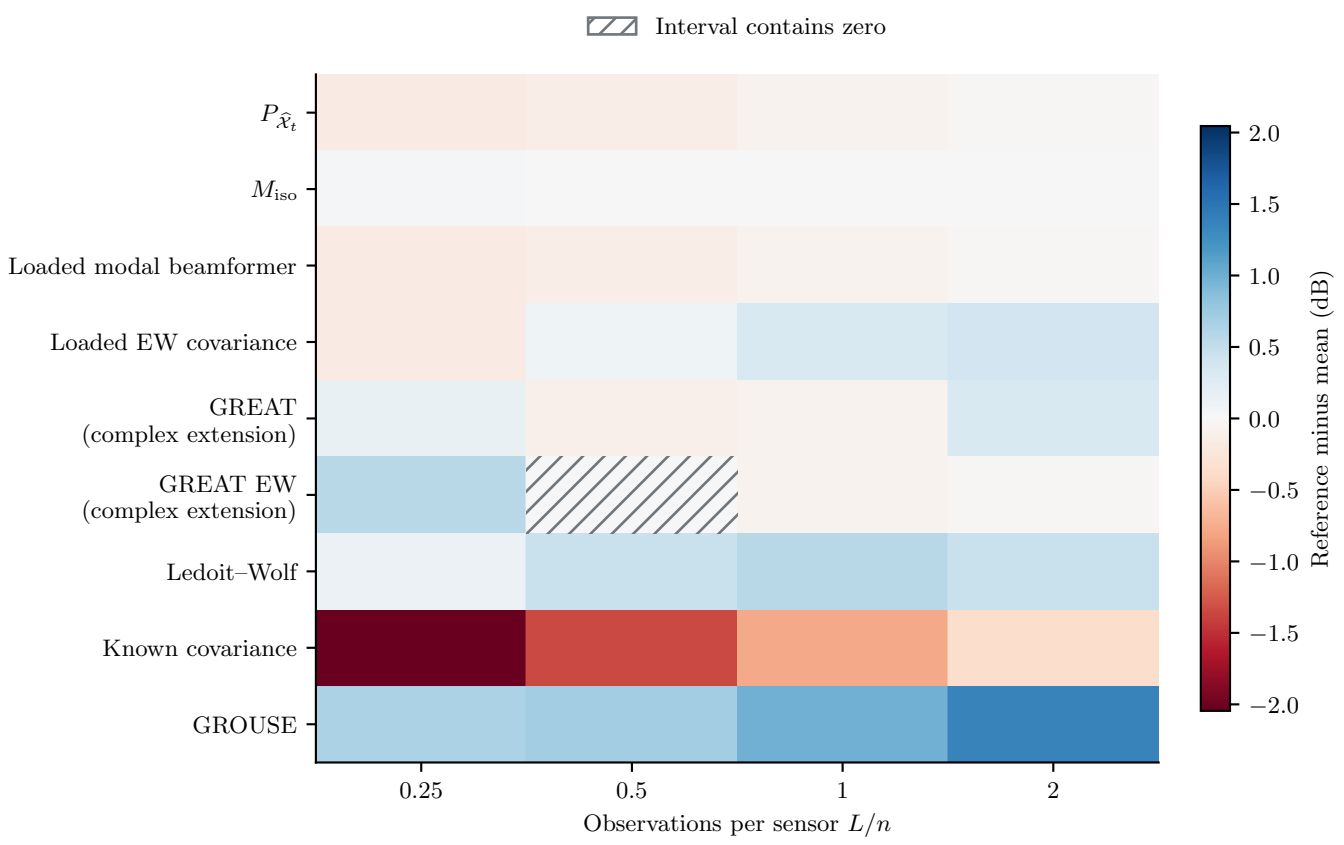
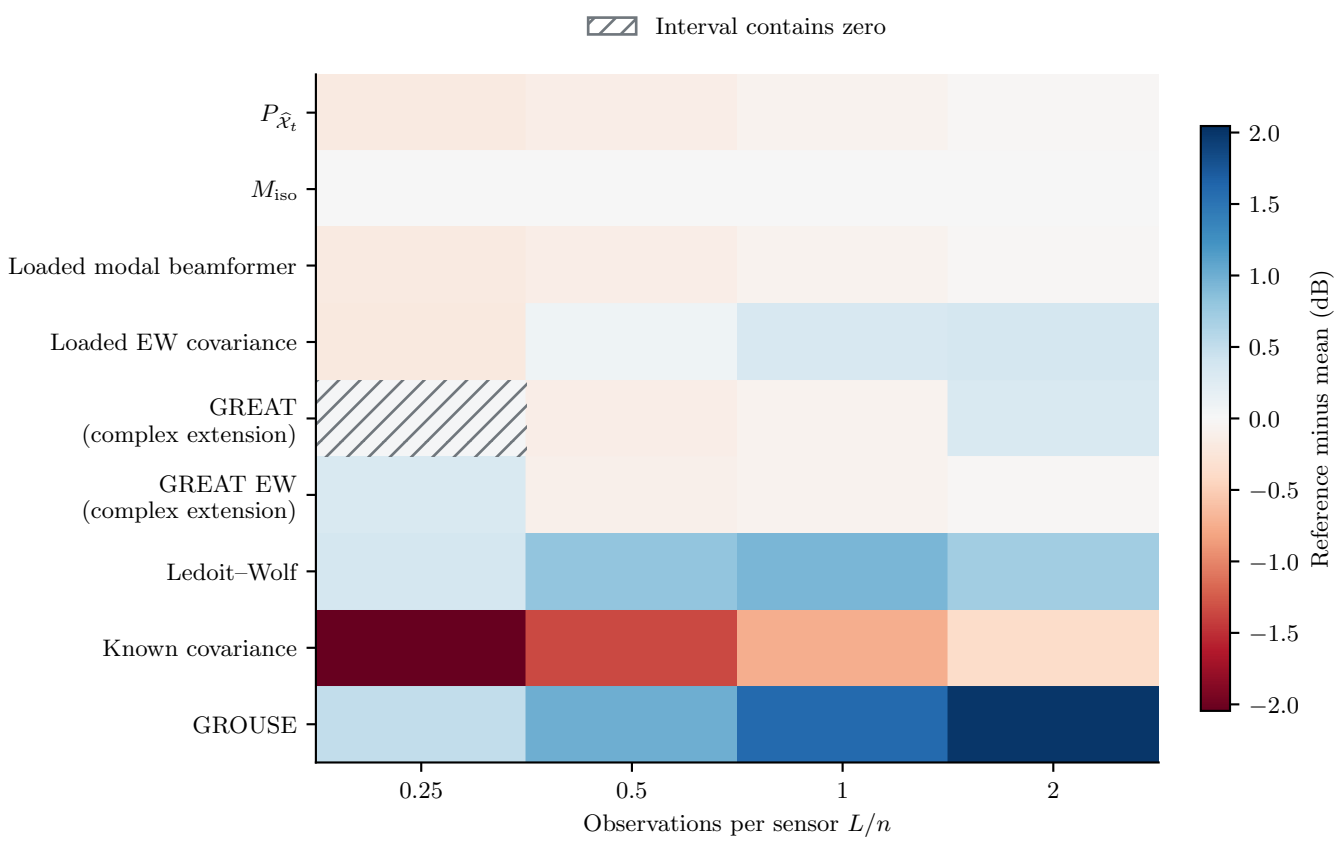
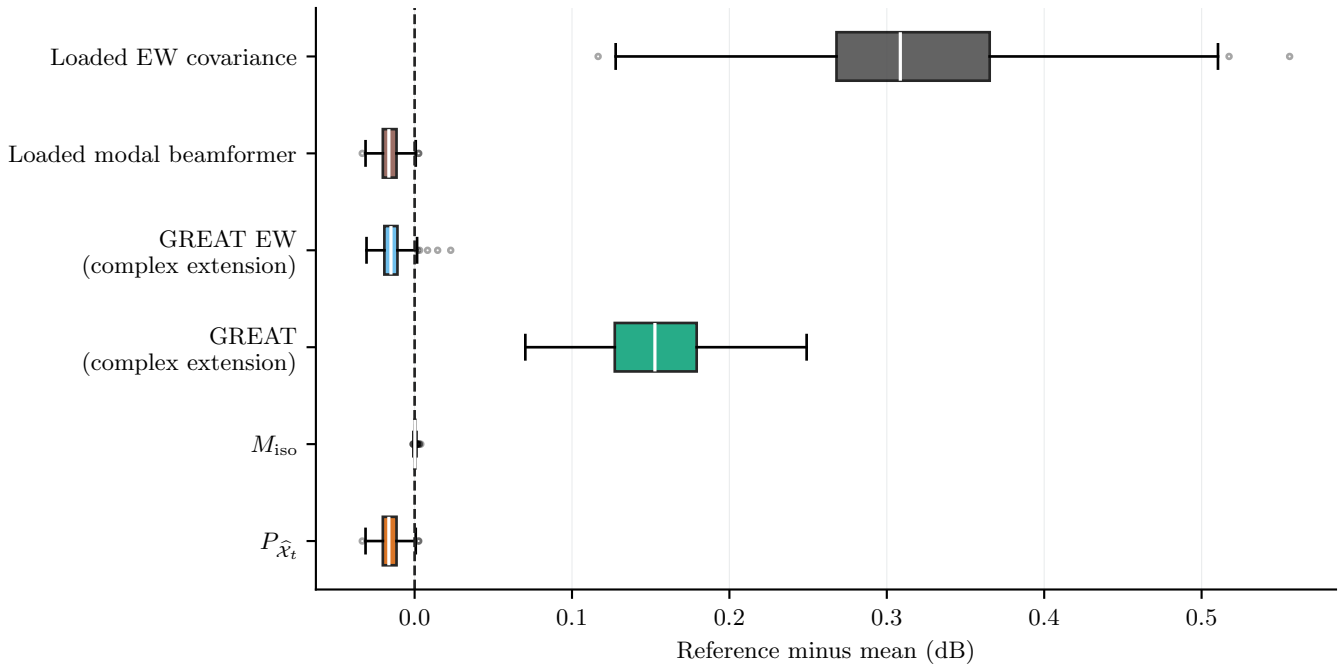

(a)
Output power / oracle power (dB)
1.1
1.0
0.9
0.8
0.7
40
60
80
100
120
Number of sensors
(b)
GREAT EW
(complex extension)
GREAT
(complex extension)
M_{iso}
$P_{\widehat{\mathcal{X}}_t}$
−0.10
−0.05
0.00
0.05
Reference minus mean (dB)
$\overline{P}_t^+$
$P_{\widehat{\mathcal{X}}_t}$
Loaded EW covariance
GREAT
(complex extension)

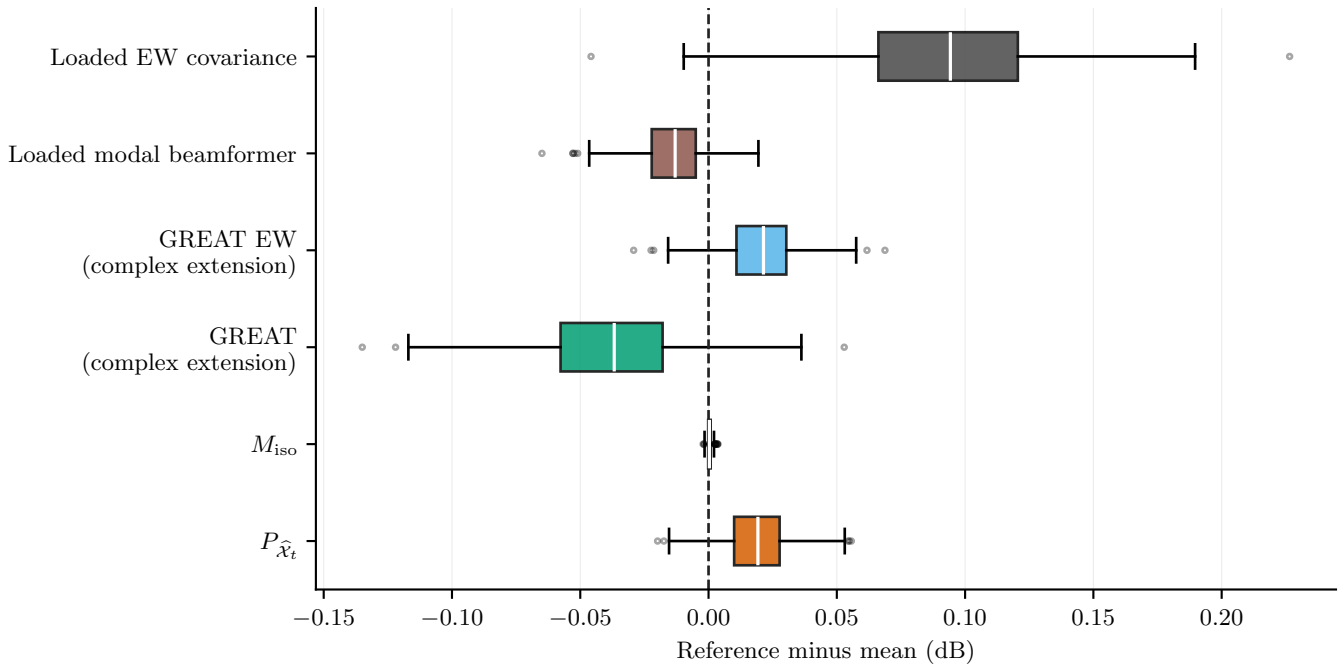

(a)
Aggregation discrepancy
10^{-10}
10^{-12}
10^{-15}
parameter
mean
mode
0
10
20
30
Observations per block
(b)
Output power / oracle power (dB)
3.0
2.5
2.0
1.5
1.0
Delayed exact aggregation
Block likelihood
0
10
20
30
Observations per block
(c)
Action delay (s)
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
0
10
20
30
Observations per block

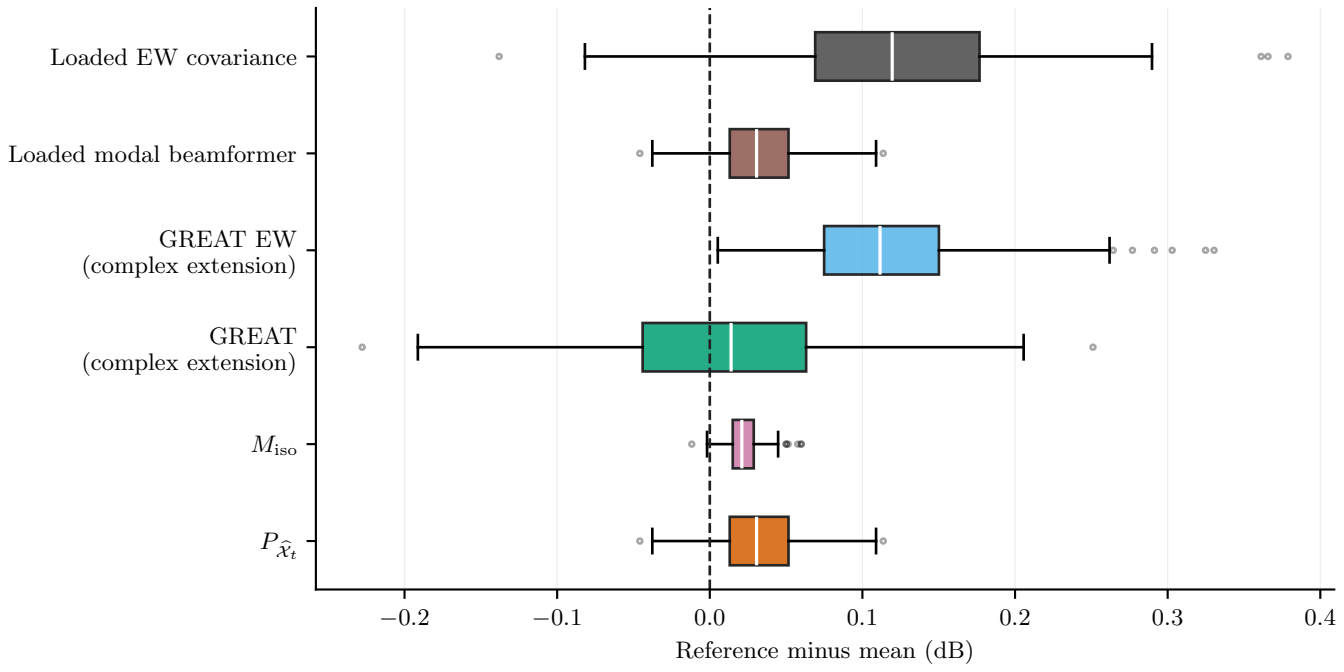

Interval contains zero
$P_{\widehat{\mathcal{X}}_t}$
M_{iso}
Loaded modal beamformer
Loaded EW covariance
GREAT (complex extension)
GREAT EW (complex extension)
Ledoit–Wolf
Known covariance
GROUSE
0.25
0.5
1
2
Observations per sensor L/n
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Reference minus mean (dB)

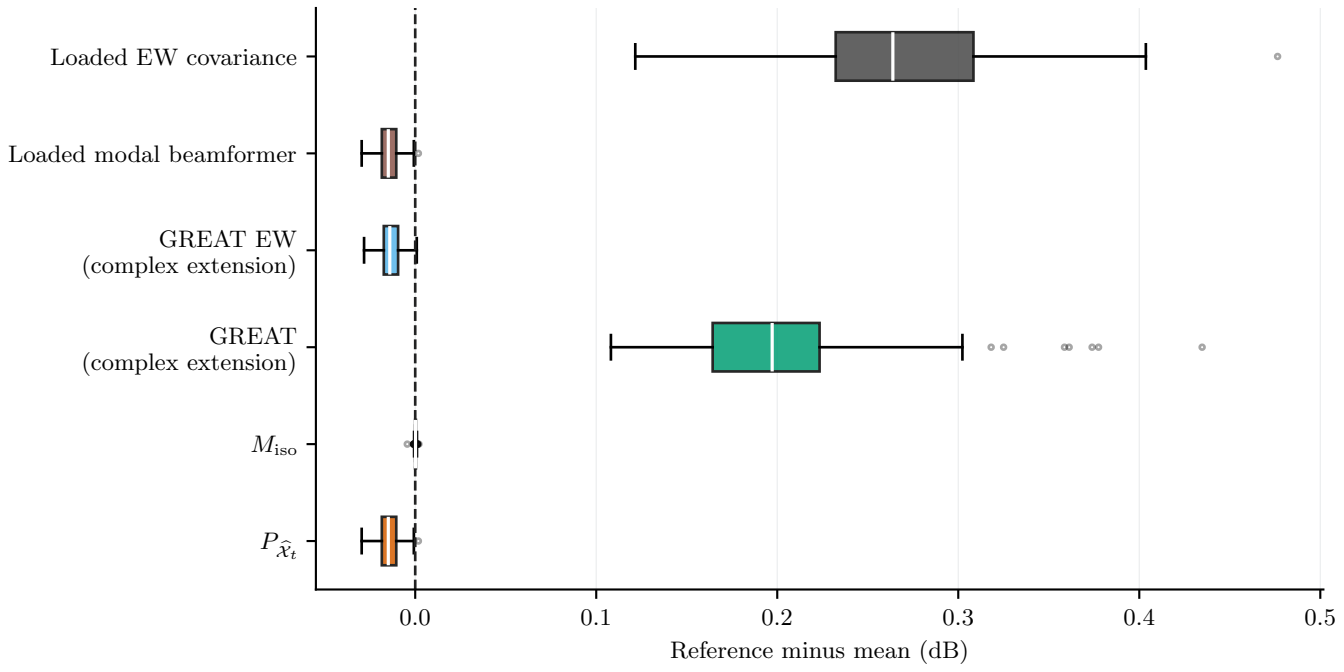

Interval contains zero
$P_{\widehat{\mathcal{X}}_t}$
M_{iso}
Loaded modal beamformer
Loaded EW covariance
GREAT (complex extension)
GREAT EW (complex extension)
Ledoit–Wolf
Known covariance
GROUSE
0.25
0.5
1
2
Observations per sensor L/n
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Reference minus mean (dB)

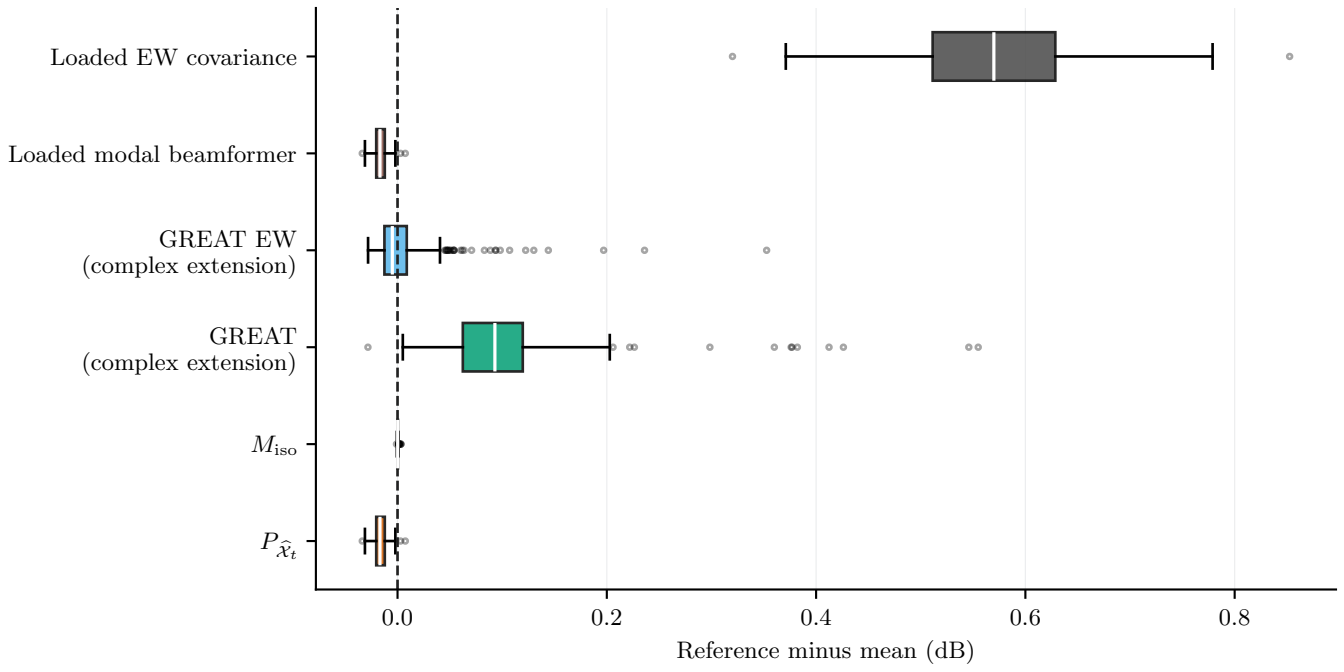

Interval contains zero
$P_{\widehat{\mathcal{X}}_t}$
M_{iso}
Loaded modal beamformer
Loaded EW covariance
GREAT
(complex extension)
GREAT EW
(complex extension)
Ledoit–Wolf
Known covariance
GROUSE
0.25
0.5
1
2
Observations per sensor L/n
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Reference minus mean (dB)

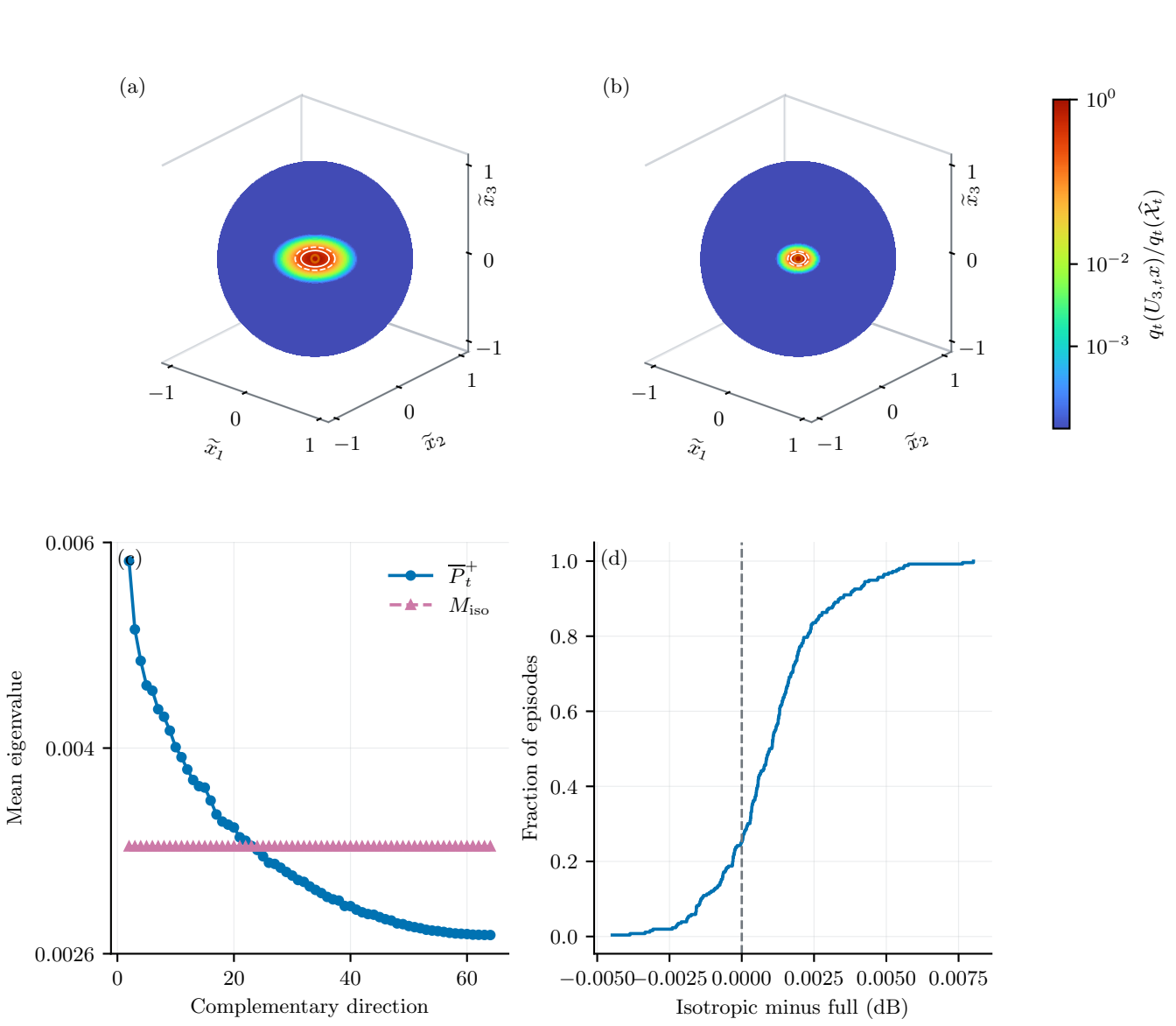

Loaded EW covariance
Loaded modal beamformer
GREAT EW
(complex extension)
GREAT
(complex extension)
M_{iso}
$P_{\widehat{\mathcal{X}}_t}$
0.0
0.1
0.2
0.3
0.4
0.5
Reference minus mean (dB)


Loaded EW covariance
Loaded modal beamformer
GREAT EW
(complex extension)
GREAT
(complex extension)
M_{iso}
$P_{\widehat{\mathcal{X}}_t}$
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
Reference minus mean (dB)


Loaded EW covariance
Loaded modal beamformer
GREAT EW
(complex extension)
GREAT
(complex extension)
M_{iso}
$P_{\widehat{\mathcal{X}}_t}$
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
Reference minus mean (dB)


Loaded EW covariance
Loaded modal beamformer
GREAT EW
(complex extension)
GREAT
(complex extension)
M_{iso}
$P_{\widehat{\mathcal{X}}_t}$
0.0
0.1
0.2
0.3
0.4
0.5
Reference minus mean (dB)


Loaded EW covariance
Loaded modal beamformer
GREAT EW
(complex extension)
GREAT
(complex extension)
M_{iso}
$P_{\widehat{\mathcal{X}}_t}$
0.0
0.2
0.4
0.6
0.8
Reference minus mean (dB)


(a)
(b)
$\widetilde{x}_3$
$\widetilde{x}_1$
$\widetilde{x}_2$
10^0
10^{-2}
10^{-3}
$q_t(U_{3,t}x)/q_t(\widehat{\mathcal{X}}_t)$
(c)
$\overline{P}_t^+$
M_{iso}
Mean eigenvalue
Complementary direction
(d)
Fraction of episodes
Isotropic minus full (dB)